



**A 15 ps Two-Dimensional Vernier
Time-to-Digital Converter in 180 nm BCD**

Design report
Digital IC Design II (EE4615)

5-bit thermometer output, calibrated resolution 15.0–15.85 ps, validated in all five process corners.
Schematic and Spectre simulation only; every cell hand-designed in TSMC 180 nm BCD.

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Delft, 15 June 2026

Abstract

This report presents the design and verification of a 5-bit two-dimensional Vernier time-to-digital converter (TDC) in TSMC 180 nm BCD. Two delay lines with $\tau_1 = 90$ ps and $\tau_2 = 75$ ps span a 10×6 comparison grid whose 31 thermometer levels each map to exactly one SR-latch arbiter, so no combining logic is needed. The resolution $t_0 = \tau_1 - \tau_2 = 15$ ps sits well below the 20 ps specification and below the technology gate delay. Three design decisions carry the result: fan-out-of-one delay cells that break the 97 ps floor of conventional fan-out-of-four sizing, characterization against a replica of the true in-grid load of roughly twenty unit inverters, and dummy latches that equalize the fan-out of every tap. With one fixed trim setting the converter passes all sign-off checks in all five process corners (LSB 13.1–18.2 ps, worst DNL 0.90 LSB, no missing codes). Switched MOSCAP trim banks then re-center the LSB per corner to 15.0–15.85 ps, a six times tighter spread. The core measures 11.7 pJ per conversion at a figure of merit of 0.50 pJ per step, with a total transistor width of 1305.6 μm . The arbiter dead zone is below 1 fs. A one-dimensional Vernier row built from the same cells was measured for comparison and confirms both the loading analysis and the area advantage of the two-dimensional architecture.

1 Introduction

1.1 Background and motivation

A time-to-digital converter quantizes the interval between two signal edges. In an all-digital PLL it replaces the analog phase detector and charge pump, which made TDCs standard in digital RF synthesis; they also appear in LiDAR and time-of-flight imaging [4, 7]. The basic flash TDC, a delay line sampled along its taps, has a resolution floor of one gate delay. In 180 nm CMOS that floor is several tens of picoseconds under realistic loading (our own loaded stage measures 97 ps at fan-out-of-four sizing), above the 20 ps this project requires, so a sub-gate-delay technique is mandatory.

The Vernier principle reaches below the gate delay by racing two delay lines: the START edge propagates through stages of delay τ_1 while the STOP edge chases it through faster stages of delay τ_2 . The effective resolution is the difference $t_0 = \tau_1 - \tau_2$, which can be made arbitrarily small in principle [3]. The classical one-dimensional implementation pays for this with hardware and latency: one stage pair and one arbiter per code, so range and resolution fight each other. Ring topologies reuse a short line to recover range at the cost of control complexity [5]; passive interpolation attacks the problem from the matching side [6]. The two-dimensional Vernier of Liscidini, Vercesi and Castello [1, 2] instead compares *every* START tap against *every* STOP tap, so a grid of $N_X + N_Y$ stages covers $\mathcal{O}(N_X N_Y)$ distinct intervals. We chose this architecture: it decouples range from line length and suits a course budget measured in transistor width.

1.2 Objectives and boundary conditions

The EE4615 brief sets hard requirements and a design-quality trade-off (Table 1). Every gate must be hand-designed in TSMC 180 nm BCD; standard cells are not allowed. Verification is by Spectre simulation of the schematic, no layout. The course testbench fixes the interface: a 40 ps RESET pulse, START and STOP inputs each driven through $3\times$ minimum inverters, and 31 thermometer outputs $Q_{1..31}$, each loaded by the TDC_out cell (two $2\times$ inverters and 1 fF) [8]. Robustness must be demonstrated in the five process corners TT, SS, FF, SF and FS [9].

Table 1: Project requirements and quality metrics.

Requirement	Value
Output	5 bit, thermometer code $Q_{1..31}$
Resolution (LSB)	better than 20 ps
Range	full 32-code linear range
Process / supply	TSMC 180 nm BCD, 1.8 V
Corners	TT, SS, FF, SF, FS, all must pass
Standard cells	none; all cells hand-designed
Verification	schematic + Spectre only
Quality metric	Definition
Linearity	DNL, INL per corner
Energy	average energy per conversion
Area	total transistor width ΣW
Figure of merit	energy per conversion step

1.3 Contributions and report structure

The delivered converter reaches a calibrated resolution of 15.0–15.85 ps in all five corners with no missing codes. Beyond meeting the specification, the project contributes three results that we believe generalize: a quantified demonstration that fan-out-of-four sizing cannot reach the required stage delays in this technology, a replica-load sizing method for Vernier grids in which each tap drives on the order of twenty unit gates, and a measured one-dimensional versus two-dimensional comparison built from identical cells that isolates the architectural benefit from the circuit design.

Section 2 develops the architecture and the operating point. Section 3 documents every cell with its transistor sizes. Section 4 describes the design iterations and the verification setup. Section 5 reports the corner, calibration, energy, metastability and comparison results, and Section 6 concludes.

2 Architecture

2.1 From one to two dimensions

In a 1-D Vernier TDC the STOP edge gains $t_0 = \tau_1 - \tau_2$ on the START edge at every stage, and an arbiter at each stage records whether STOP has caught up. Resolving the 31 thermometer levels therefore takes 31 stages in each line and 31 arbiters, and the conversion latency grows with the full $31 \tau_1$. The 2-D Vernier [2] generalizes the race: with START applied to a line of N_X stages of delay τ_1 and STOP to a line of N_Y stages of delay τ_2 , an arbiter at grid position (i, j) compares the edge leaving START stage i with the edge leaving STOP stage j and fires when

$$\Delta t > i \tau_1 - j \tau_2. \quad (1)$$

Each cell therefore measures its own threshold, and $N_X + N_Y$ stages yield up to $N_X N_Y$ distinct thresholds. Range no longer dictates line length.

2.2 Bijective level mapping at $k = 6$

We restrict the design space to the family $\tau_1 = k t_0$, $\tau_2 = (k - 1) t_0$. In this family the threshold of cell (i, j) equals $(ik - j(k - 1)) t_0$, an exact integer multiple of the LSB, and the mapping from thermometer level m to grid cell is bijective [2]:

$$y_m = ((m - 1) \bmod k) + 1, \quad x_m = \frac{m + (k - 1) y_m}{k}, \quad m = 1, \dots, 31. \quad (2)$$

Level m is exactly one SR latch, wired straight to output Q_m . No OR tree or encoder sits between arbiter and output, so combining-logic skew cannot corrupt the code. For $k = 6$ the mapping spans $x_m \leq 10$ and $y_m \leq 6$: a 10×6 logical grid, 10 START stages and 6 STOP stages for all 31 levels.

The operating point follows from two constraints. The LSB must clear the 20 ps specification with margin for corner spread, which sets $t_0 = 15$ ps. The stage delays must stay inside the window that the delay cell can actually produce under its in-grid load (Section 4), which selects $k = 6$: $\tau_1 = 90$ ps and $\tau_2 = 75$ ps. Smaller k would require τ_2 below the achievable minimum; larger k would lengthen both lines and add area without improving resolution.

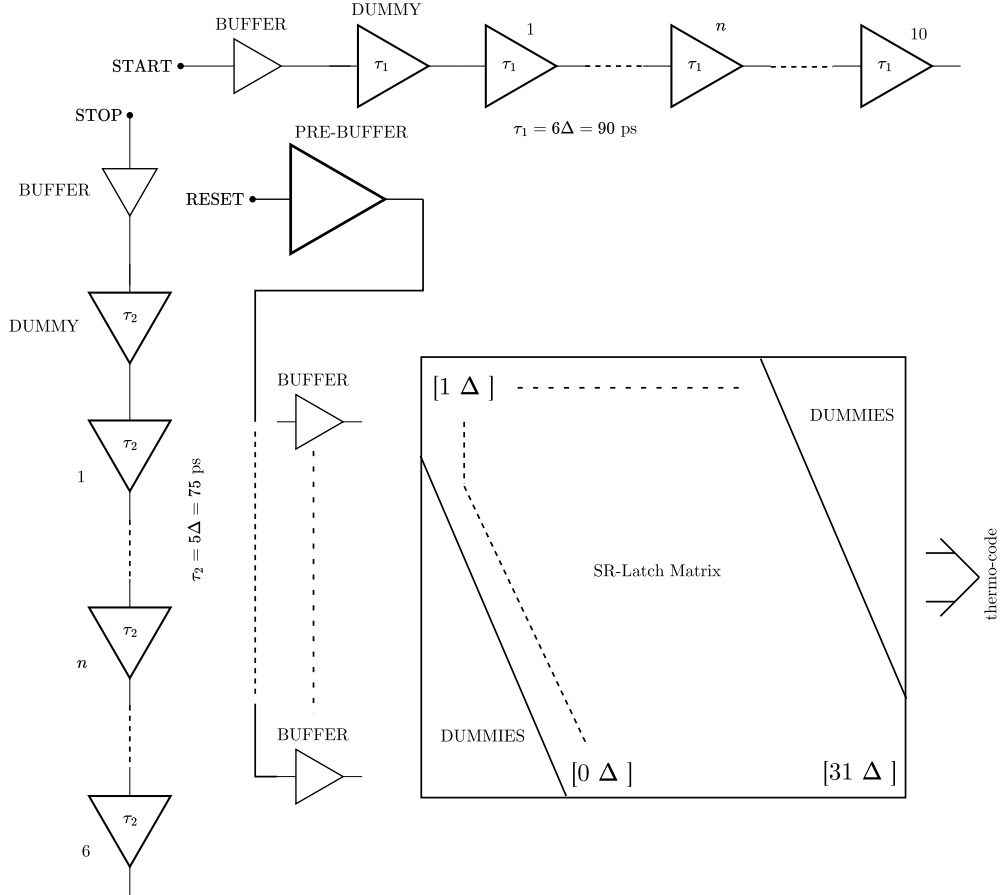


Figure 1: Top level of the converter. The START line (10 stages of τ_1 , top) and the STOP line (6 stages of τ_2 , left) feed the SR-latch matrix. The 31 active latches sit on the bijective map of Eq. (2); dummy latches pad every tap to identical fan-out. Buffer trees distribute START, STOP and the 40 ps RESET. The thermometer code is read directly from the latch outputs.

2.3 Fan-out equalization and reset

Equation (1) only holds if every stage of a line contributes the same delay, and stage delay depends on load. In the raw 10×6 grid the taps drive between one and ten latches, which would modulate the stage delays along the line and corrupt the thresholds. We therefore pad the array with 29 dummy latches to a physically symmetric 10×10 arrangement: 31 active plus 29 dummy latches give every START tap and every STOP tap the same fan-out of ten latch inputs. The dummies are electrically identical to active latches; only their outputs are unused. The same uniformity argument adds one extra delay cell at the head of each line (11 and 7 cells in total), so that the first logical stage sees the same driver as all others.

Each latch carries an asynchronous clear input. The 40 ps RESET pulse from the testbench,

widened through its own buffer tree, clears all 60 latches before each conversion so that every conversion starts from $Q_m = 0$.

3 Circuit design

All cells were designed at transistor level in the `vernier2d` library using the 2 V devices of the `tsmc18` library (`nmos2v/pmos2v`, Spectre models `nch/pch`) at the minimum channel length $L = 180$ nm. The unit inverter used throughout is $W_p/W_n = 440$ nm/220 nm; larger drivers are built as parallel fingers (multiples) of this unit. Table 2 summarizes the cell inventory; the per-cell width budget is given in Appendix B.

Table 2: Cell inventory of the TDC core with device sizing. All lengths 180 nm except the MOSCAPs (250 nm).

Cell	Count	Contents and sizing
<code>delay_tau1</code>	11	FO=1 inverter pair ($5\times$ input, $6\times$ driver, unit 440/220 nm); fixed MOSCAP ($18\times$ fingers) behind an always-on access device ($7\times$); three TG-switched trim banks of 2/20/21 fingers.
<code>delay_tau2</code>	7	identical active devices as <code>delay_tau1</code> ; lighter trim banks of 2/14/18 fingers.
<code>srlatch</code>	60	13 transistors: two 2-input NANDs (all devices 440/180 nm), asynchronous-reset NMOS ($2.0/0.18\ \mu\text{m}$), two unit-inverter output buffers.
<code>nand</code> <code>Inv_10x/20x/40x/80x</code>	(in latch) trees	two series NMOS, two parallel PMOS, all 440/180 nm. input and reset distribution buffers, parallel-finger multiples of the unit inverter.

3.1 Delay stage

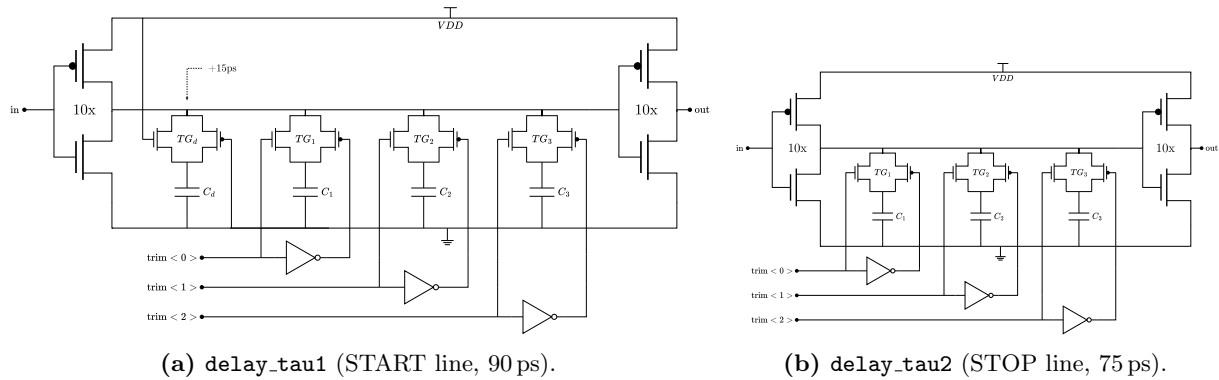


Figure 2: Delay stages. An FO=1 inverter pair drives the next stage; the internal node carries one fixed MOSCAP behind an always-on access transistor plus three transmission-gate-switched MOSCAP banks controlled by `trim<0:2>`. The two cells differ only in bank sizes.

The delay stage is a two-inverter buffer whose internal node is loaded capacitively (Fig. 2). Two choices define it.

Fan-out of one. Conventional FO=4 sizing minimizes the delay of a logic path, but here delay is the product, not the cost. With FO=4 the intrinsic delay of the internal node pins the minimum achievable stage delay at 97 ps in this technology, which already exceeds the required τ_2 . Equal input and driver sizing ($5\times$ and $6\times$ unit fingers) reduces the internal-node capacitance and opens a usable window of 74–147 ps under the true grid load.

Capacitive tuning, identical actives. Both lines use the same active devices; only the capacitance on the internal node differs. The fixed MOSCAP ($18\times$ fingers, $L = 250$ nm) sets the nominal delay, and three binary-weighted-like banks ($2/20/21$ fingers on the τ_1 cell, $2/14/18$ on the τ_2 cell) are switched in or out through transmission gates by the static `trim<0:2>` inputs. Because the 6:5 delay ratio comes from loading rather than device sizing, τ_1 and τ_2 track each other over process and temperature, which is what keeps their difference, the LSB, stable. At the tuned typical-corner point the lines measure $\tau_1 = 90.5$ ps and $\tau_2 = 75.4$ ps.

3.2 Arbiter

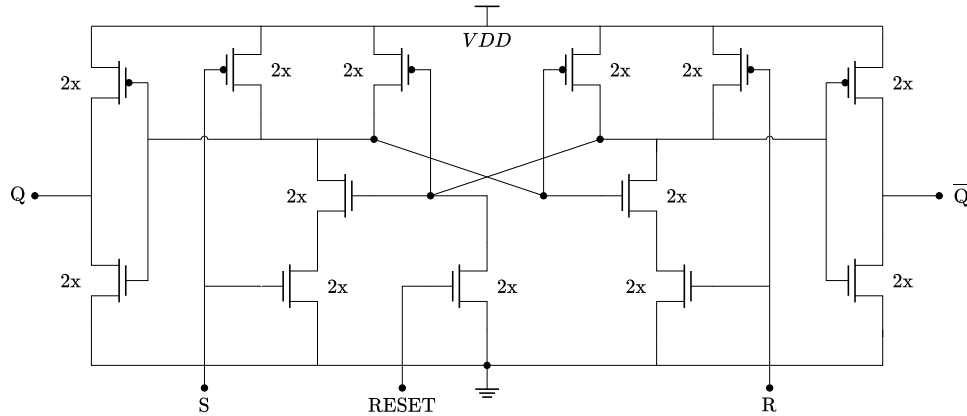


Figure 3: SR-latch arbiter (`vernier2d/srlatch`). Two cross-coupled NANDs (440/180 nm throughout), an asynchronous-reset NMOS on the internal node, and two unit-inverter output buffers. S and R are active high; the first rising edge wins.

The arbiter decides, at each grid node, which edge arrived first. We use a cross-coupled NAND SR latch with buffered outputs (Fig. 3), 13 transistors in total. Both NAND outputs idle high; the first rising input flips the latch and positive feedback locks the decision. Three properties motivated the choice over a D flip-flop. The cell is small, which matters at 60 instances. The S and R paths are electrically symmetric with equal NMOS and PMOS widths, which is what keeps the arbitration offset at the femtosecond level measured in Section 5.6. And reset costs a single NMOS ($2.0/0.18\ \mu\text{m}$) that pulls one internal node during the 40 ps RESET pulse, defining $Q = 0$ before every conversion.

3.3 Core assembly

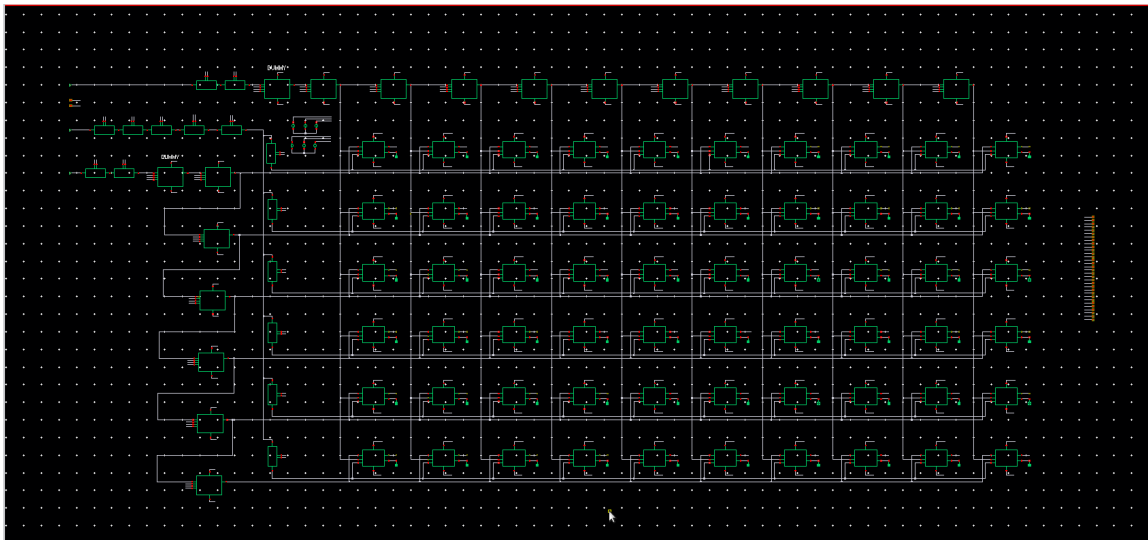


Figure 4: `vernier2d/tdc_core` as entered in Cadence Virtuoso: START line across the top, STOP line down the left, and the 10×10 padded latch matrix. The core contains 1120 active devices.

The core (Fig. 4) instantiates 11 `delay_tau1` and 7 `delay_tau2` stages, 60 latches and the input buffer trees, and exposes the course-mandated pins (RESET, START, STOP, V_{DD} , GND, $Q_{1..31}$). The six trim lines enter as static gate biases, standing in for configuration bits. The total transistor width is $\Sigma W = 1305.6 \mu\text{m}$: $1078.3 \mu\text{m}$ of active devices (1120 transistors) and $227.3 \mu\text{m}$ of MOSCAP (65 devices). The width budget is dominated by the `delay_tau1` drivers ($451 \mu\text{m}$) and the latch NANDs ($211 \mu\text{m}$); the full breakdown is in Appendix B.

4 Design methodology and verification setup

4.1 Sizing against the real load

The decisive step of the design was recognizing what a delay stage actually drives inside the grid. A τ_2 tap feeds the latch inputs of an entire row, a τ_1 tap those of an entire column, and both feed the next delay stage, whose input capacitance scales with the stage sizing itself. After fan-out equalization the equivalent load is about twenty unit-inverter inputs per tap.

We learned the cost of ignoring this directly. Stage sizing derived from a standalone sweep with an idealized load predicted $t_0 = 16.6 \text{ ps}$; the same cells assembled into a row with raw tap loading measured 28.6 ps (Section 5.7). Nearly a factor of two in resolution, lost entirely to loading. All final sizing was therefore done against a replica load of ten latch inputs plus one downstream stage, swept in the slowest corner (SS) so that the slow line stays reachable everywhere. This analysis also moved the architecture itself: an earlier $k = 4$ operating point ($\tau_1/\tau_2 = 60/45 \text{ ps}$) required stage delays below the achievable window and was abandoned in favor of $k = 6$.

4.2 Per-corner calibration

Process corners shift both line delays together, but not perfectly, so t_0 breathes with the corner. The MOSCAP banks of Section 3.1 absorb this. The calibration table was generated by an exhaustive probe: all 64 on/off combinations of the six bank gates were simulated per corner (320 short transients), measuring τ_1 and τ_2 directly on the tap nets. The configuration with t_0 closest to 15 ps wins; ties resolve toward τ_1 closest to 90 ps . The winning codes (Appendix A) were then validated end-to-end with full staircase sweeps. An independent hand derivation of the

per-corner codes agreed within one bank step in every corner. Calibration changes gate voltages only, so the transistor inventory and ΣW are identical in every corner.

4.3 Testbench and measurement method

All system-level results come from the unmodified course testbench TB.TDC [8]: stimulus generation in TDC_in (RESET, START, STOP through $3\times$ minimum inverters), the DUT wrapper, and TDC_out loading every thermometer output with two $2\times$ inverters and 1 fF. The DUT receives its supply through a dedicated rail with a series ammeter, while the stimulus buffers run from a separate testbench rail, so the measured energy contains only the converter. Energy per conversion integrates the supply current over each 5 ns conversion window, $E = V_{DD} \int |i| dt$.

The characterization is automated in OCEAN. Per corner, the input delay is swept from -15 ps in 1.5 ps steps (one tenth of the LSB) until code 31 is reached, recording every code transition; the sweep then continues about 15% further for the energy average. From the transition table the scripts extract the LSB (mean step width), DNL, INL, offset, energy, ENOB and the sign-off checks: no missing codes, monotonicity, and an intact thermometer code at every point. The five corners run as parallel, isolated Spectre jobs (1670 transients in the corner campaign). Two INL conventions appear in the outputs: the course script reports the half-range value $(INL_{\max} - INL_{\min})/2$, while the plots show the endpoint-referenced per-code curve. We quote the course convention for sign-off and plot the endpoint curve, stating both where they differ.

5 Results

All results are Spectre simulations of the schematic at 27°C and $V_{DD} = 1.8\text{ V}$ through the course testbench, in the five corners TT, SS, FF, SF and FS.

5.1 Transfer and resolution across corners

Figure 5 shows the measured transfer with one fixed trim configuration (the tuned TT setting) in all five corners; Table 3 lists the extracted metrics. Every corner produces a monotonic 31-step staircase with an intact thermometer code, and every corner passes the 20 ps requirement, with the slowest corner (SS, 18.2 ps) closest to the limit. The skewed corners SF and FS barely differ from TT because opposite N and P shifts largely cancel in an inverter pair. The input-referred offset stays between -7 and -24 ps, constant per corner, and is a code offset rather than a linearity error.

Table 3: Corner results with one fixed trim configuration. INL is the course half-range convention; energy and FoM from the supply-ammeter measurement. Every corner passes all sign-off checks (no missing codes, monotonic, LSB below 20 ps).

Corner	LSB (ps)	DNL _{max} (LSB)	INL (LSB)	Offset (ps)	E/conv (pJ)	ENOB	FoM (pJ/step)
TT	15.35	0.66	0.65	-15.4	11.67	4.54	0.50
SS	18.20	0.90	0.76	-24.2	10.76	4.39	0.51
FF	13.10	0.72	0.58	-7.1	12.84	4.58	0.54
SF	15.35	0.66	0.61	-13.9	11.96	4.57	0.50
FS	15.70	0.72	0.61	-15.7	11.38	4.56	0.48

5.2 Linearity

Worst-case DNL is 0.90 LSB (SS), well clear of the -1 LSB missing-code limit; no corner loses a code (Fig. 6). Switching from the half-range convention of Table 3 to the per-code endpoint curve of Fig. 6b, the worst endpoint INL is 1.04 LSB (SS); all other corners stay at or below

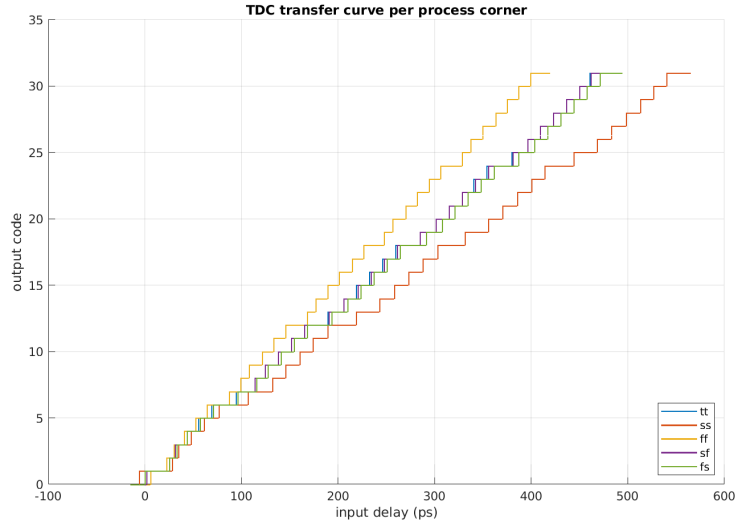
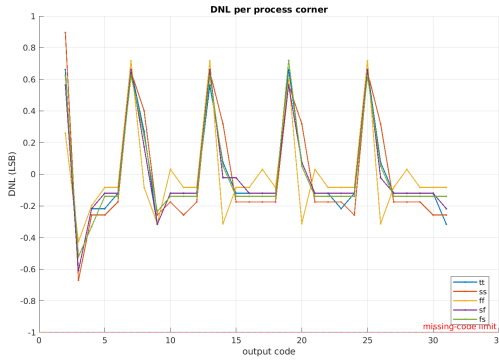
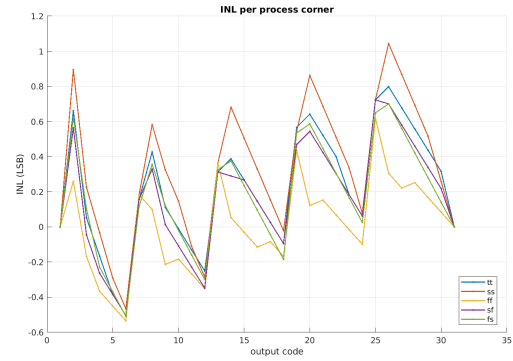


Figure 5: TDC transfer per corner, fixed trim configuration. All corners are monotonic and reach code 31; the slope spreads with corner speed.



(a) DNL per corner.



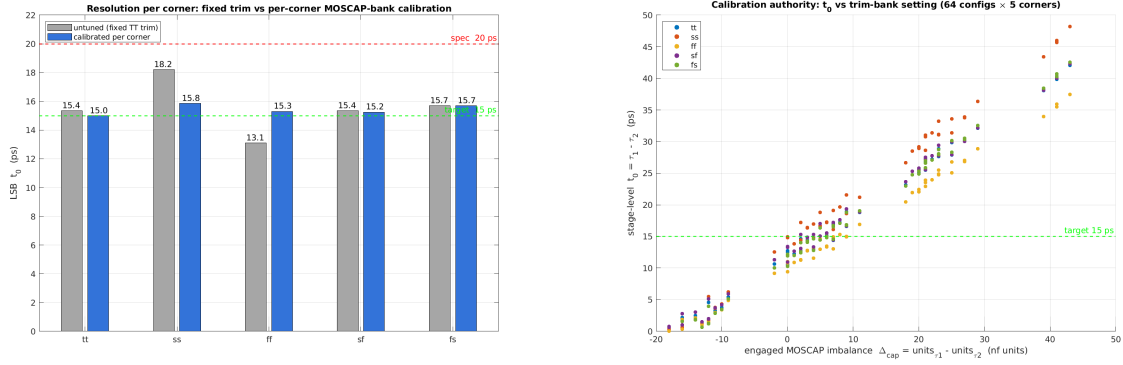
(b) INL per corner (endpoint convention).

Figure 6: Linearity with fixed trim. The period-6 sawtooth marks the column-wrap boundaries of the $k = 6$ mapping and repeats identically in every corner.

0.80. The dominant DNL feature is a systematic sawtooth with period 6: peaks of 0.6–0.7 LSB at codes 7, 13, 19 and 25, exactly the column-wrap boundaries of the mapping in Eq. (2), where the threshold hands over from a row step to a column step. The pattern is identical in every corner, which identifies it as an architectural path-length imbalance at the wrap rather than device mismatch. Calibration does not remove it (Section 5.3); a targeted per-wrap capacitive correction could, and is listed as future work.

5.3 Per-corner calibration

Applying the per-corner trim codes of Appendix A re-centers the LSB to 15.0–15.85 ps: the corner spread tightens from 5.10 ps to 0.85 ps, six times, and the worst-case margin to the specification grows from 1.8 to 4.15 ps (Fig. 7a). DNL and INL barely move (worst tuned INL 0.74 LSB), as expected: calibration re-centers the average step but cannot touch the architectural wrap pattern. One honest data point: the FS optimum is the default configuration, since that corner shifts N and P devices oppositely and nearly cancels. All sign-off checks still pass in every corner.



(a) LSB per corner, fixed trim vs calibrated.

(b) Calibration authority: stage-level t_0 vs engaged capacitance imbalance, all 64 configurations in 5 corners.

Figure 7: MOSCAP-bank calibration. Every corner re-centers to the 15 ps target; the banks command roughly 0–48 ps, far more authority than the corners require.

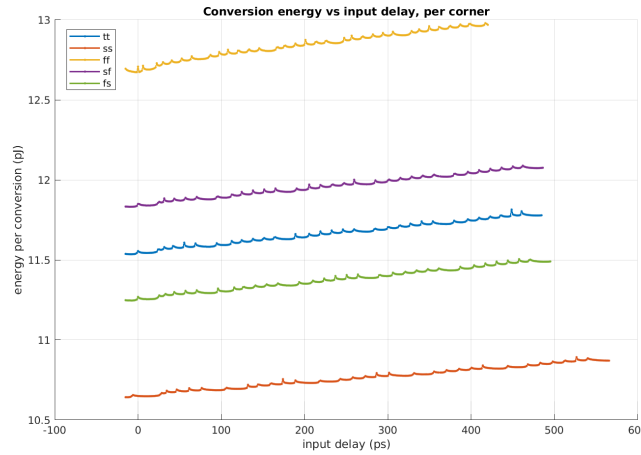


Figure 8: Energy per conversion vs input delay, per corner. The band is nearly flat; energy tracks corner speed.

5.4 Energy, power and figure of merit

Energy per conversion sits in a flat 10.6–13.0 pJ band across input delay and corner (Fig. 8), ordered by corner speed (SS lowest, FF highest). With the 5 ns conversion window of the testbench this corresponds to an average power of 2.15–2.57 mW. Table 4 reports both figure-of-merit conventions: the specification's P/t_0 and the course script's $E/2^{\text{ENOB}}$. The dynamic range, $31 \times \text{LSB}$, spans 406–564 ps depending on the corner, with a quantization-limited single-shot precision $\text{LSB}/\sqrt{12}$ of 3.8–5.3 ps.

Table 4: Power and figure of merit per corner (fixed trim). P_{avg} assumes the 5 ns conversion window (200 MS/s).

Corner	P_{avg} (mW)	P/t_0 (mW/ps)	$E/2^{\text{ENOB}}$ (pJ/step)
TT (nominal)	2.33	0.152	0.50
SS	2.15	0.118	0.51
FF	2.57	0.196	0.54
SF	2.39	0.156	0.50
FS	2.28	0.145	0.48

5.5 Area

The transistor-width total of the core is $\Sigma W = 1305.6 \mu\text{m}$: $1078.3 \mu\text{m}$ of active width over 1120 devices plus $227.3 \mu\text{m}$ of MOSCAP over 65 devices, extracted from the Spectre netlist. Because calibration only changes gate biases, this number is identical in all five corners. Including the testbench interface cells would add $93.7 \mu\text{m}$ (TDC_{in} $11.9 \mu\text{m}$, TDC_{out} $81.8 \mu\text{m}$). The per-cell breakdown is given in Appendix B.

5.6 Arbiter dead zone and metastability

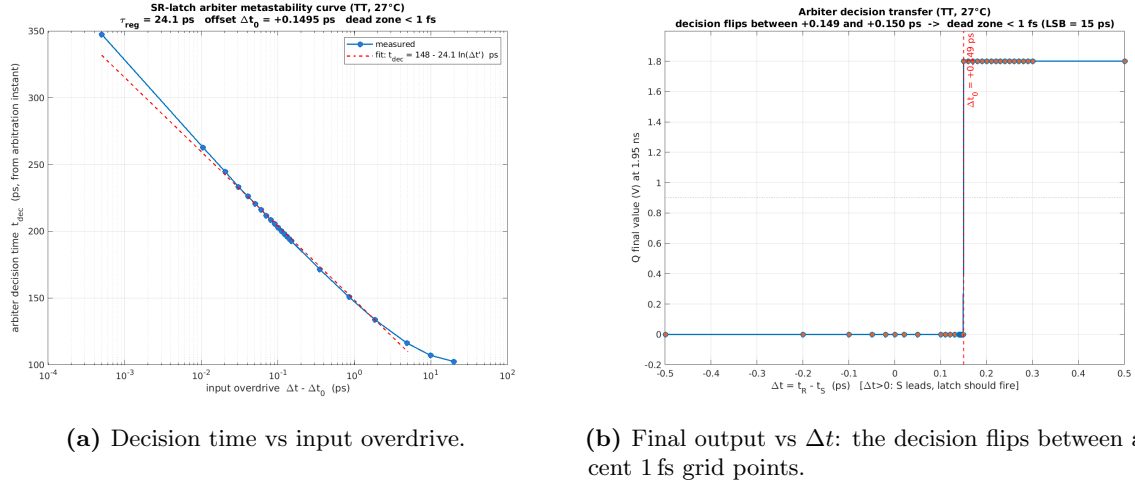


Figure 9: SR-latch arbiter characterization (TT, 27 °C, 5 fF loads), 49 simulations in three nested sweeps.

The smallest resolvable interval of the converter is set by the arbiter, so we characterized it down to 0.5 fs of input overdrive (Fig. 9). The arbitration offset is $\Delta t_0 = +149.5 \pm 0.5$ fs, about 1% of one LSB. The decision time follows the classic regeneration law $t_{dec} = t_{cq} + \tau_{reg} \ln(1/\Delta t')$ over three and a half decades with $\tau_{reg} = 24.1$ ps and a large-overdrive clk-to-Q of 102 ps; the worst observed decision time is 347.5 ps at 0.5 fs overdrive, still far inside the conversion window. No ambiguous final value was observed anywhere on the 1 fs grid, bounding the dead zone below 1 fs, four orders of magnitude under the LSB. Two caveats apply: the bound is grid-limited, and the run contains no noise or mismatch, so a Monte-Carlo analysis would widen it. Both qualifications leave orders of magnitude of margin.

5.7 Measured comparison with a 1-D Vernier

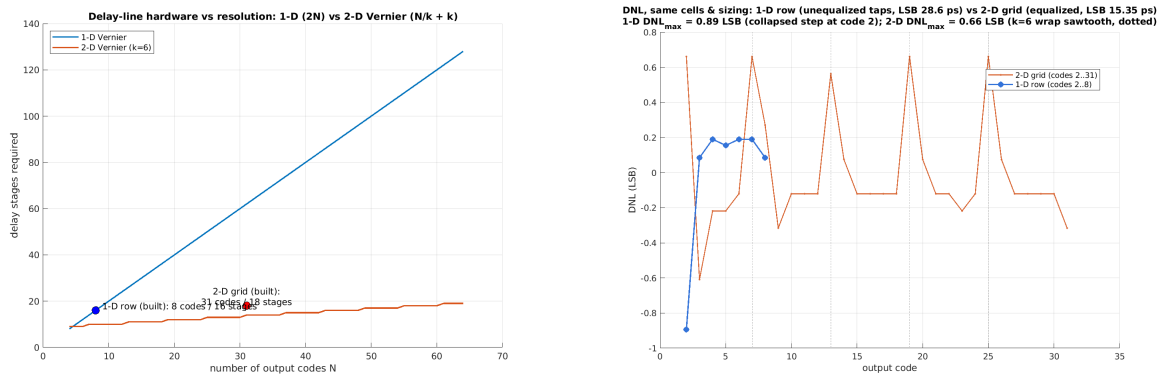


Figure 10: Architecture comparison from identical cells (TT).

To separate the architecture from the circuit design we assembled an 8-code 1-D Vernier row (`vernier2d/vernier1d_row`) from the very same delay and latch cells, with trim banks hardwired on and, deliberately, no fan-out equalization. Table 5 compares the measurements. The row resolves 28.6 ps where the stage-level prediction was 16.6 ps: direct experimental confirmation that unequalized tap loading roughly doubles the LSB, the effect that drove the sizing methodology of Section 4.1. Per code, the 2-D core is about twice as area-efficient at comparable energy. Extrapolating the row to the full 31 codes would require 62 delay stages ($3.4\times$), $\Sigma W \approx 2524\mu\text{m}$ ($1.9\times$) and a full-scale latency of 3.3 ns against 0.9 ns for the grid.

Table 5: Measured 1-D row vs 2-D core, identical cells (TT, 27 °C).

	1-D row (built)	2-D core (built)
Codes	8	31
LSB	28.6 ps	15.35 ps
DNL _{max}	0.89 LSB	0.66 LSB
Delay stages	16	18
ΣW per code	81.4 μm	42.1 μm
Energy per code	0.33 pJ	0.38 pJ
Full-scale latency (31 codes)	3.3 ns	0.9 ns

6 Conclusion

We built and verified a 5-bit two-dimensional Vernier TDC in TSMC 180 nm BCD that meets every project requirement with margin. With a single fixed trim setting the converter resolves 13.1–18.2 ps across all five process corners with no missing codes (worst DNL 0.90 LSB); per-corner MOSCAP calibration re-centers the LSB to 15.0–15.85 ps, a six times tighter spread, without changing a single device. The core consumes 11.7 pJ per conversion in the typical corner at a figure of merit of 0.50 pJ per step, within a total transistor width of 1305.6 μm . The arbiter contributes an offset of 1% of an LSB and a dead zone bounded below 1 fs, so the resolution is set by the architecture, not by the decision element.

The evidence supports three conclusions beyond the headline numbers. First, delay-cell sizing for Vernier grids must be done against the in-grid load; the measured 1-D row shows the LSB doubling when tap loading is ignored. Second, fan-out equalization with dummy latches is what makes the grid thresholds usable; it costs 29 inactive cells and buys corner-independent linearity. Third, the bijective $k = 6$ mapping removes all combining logic, and the only systematic linearity signature left, the period-6 wrap sawtooth, is architectural and identical in every corner.

Recommendations. Three extensions would add the most value. A DLL-style background loop could set the trim codes automatically, replacing the static calibration table. A voltage-to-time front end would let the TDC quantize voltages directly, addressing the course bonus. And a targeted capacitive correction at the four column-wrap codes could suppress the period-6 DNL pattern that calibration cannot reach. Layout and extracted re-simulation remain the natural next step toward silicon; the schematic-level margins reported here (4 ps on resolution, four orders of magnitude on the dead zone) leave realistic room for parasitic degradation.

References

- [1] A. Liscidini, L. Vercesi, and R. Castello, “Time to digital converter based on a 2-dimensions Vernier architecture,” in *Proc. IEEE Custom Integrated Circuits Conference (CICC)*, 2009, pp. 45–48.
- [2] L. Vercesi, A. Liscidini, and R. Castello, “Two-dimensions Vernier time-to-digital converter,” *IEEE J. Solid-State Circuits*, vol. 45, no. 8, pp. 1504–1512, Aug. 2010.

- [3] P. Dudek, S. Szczepański, and J. V. Hatfield, “A high-resolution CMOS time-to-digital converter utilizing a Vernier delay line,” *IEEE J. Solid-State Circuits*, vol. 35, no. 2, pp. 240–247, Feb. 2000.
- [4] R. B. Staszewski, S. Vemulapalli, P. Vallur, J. Wallberg, and P. T. Balsara, “Time-to-digital converter for RF frequency synthesis in 90 nm CMOS,” in *Proc. IEEE Radio Frequency Integrated Circuits (RFIC) Symposium*, 2005, pp. 473–476.
- [5] J. Yu, F. F. Dai, and R. C. Jaeger, “A 12-bit Vernier ring time-to-digital converter in 0.13 μm CMOS technology,” *IEEE J. Solid-State Circuits*, vol. 45, no. 4, pp. 830–842, Apr. 2010.
- [6] S. Henzler *et al.*, “A local passive time interpolation concept for variation-tolerant high-resolution time-to-digital conversion,” *IEEE J. Solid-State Circuits*, vol. 43, no. 7, pp. 1666–1676, Jul. 2008.
- [7] S. Henzler, *Time-to-Digital Converters*, Springer Series in Advanced Microelectronics, vol. 29, Springer, 2010.
- [8] EE4615 course staff, *EE4615 Testbench Instructions*, Delft University of Technology, Apr. 2026.
- [9] EE4615 course staff, *EE4615 Corner Analysis Manual*, Delft University of Technology, Apr. 2026.

A Per-corner trim configurations

Calibration sets the six trim-bank gates (`trim<0:2>` of each line) to 0 V or 1.8 V. Table 6 lists the winning configuration per corner from the exhaustive 64-point probe, together with the stage delays it produces. The FS optimum coincides with the default setting.

Table 6: Chosen trim-bank gate voltages per corner and resulting stage-level delays. Bank sizes: 2/20/21 fingers (τ_1), 2/14/18 fingers (τ_2).

Corner	τ_1 banks (V)			τ_2 banks (V)			τ_1 (ps)	τ_2 (ps)	t_0 (ps)
	0	1	2	0	1	2			
TT	0	1.8	0	0	1.8	0	88.5	73.4	15.04
SS	1.8	0	0	1.8	0	0	93.0	78.0	14.91
FF	1.8	1.8	1.8	1.8	1.8	1.8	90.5	75.6	14.97
SF	0	0	1.8	1.8	1.8	0	89.8	74.7	15.07
FS	1.8	1.8	0	1.8	1.8	0	92.6	77.6	15.01

The stage-level probe slightly underestimates the staircase LSB (SS: 14.91 ps probe vs 15.85 ps measured) because grid loading beyond the first stage adds a corner-dependent few percent. This is why every configuration was re-validated with a full staircase sweep.

B Transistor-width budget

Table 7 breaks the netlist-extracted ΣW of `tdc_core` down per cell, summed over all instantiations.

Table 7: ΣW breakdown of the core (Spectre netlist, tuned variable values). MOSCAPs identified by drain, source and bulk tied together. The grand total is quoted as 1305.6 μm in the text.

Group	ΣW (μm)
delay_tau1 (11 inst.)	450.89
delay_tau2 (7 inst.)	207.34
nand (latch gates)	211.20
srlatch (reset devices)	120.00
Inv (latch output buffers)	114.84
Inv_20x	92.40
Inv_80x	52.80
Inv_40x	26.40
Inv_10x	19.80
Inv_5x	9.90
Active total (1120 devices)	1078.32
MOSCAP trim banks (65 devices)	227.25
Grand total (1185 devices)	1305.57

C Simulation inventory

All results are reproducible from the OCEAN scripts submitted with the design.

- **Corner campaign** (tdc_corners_all.ocn): five corners, 1.5 ps delay step, 1670 transients; produces the transition tables behind Figs. 5–6 and Table 3.
- **Calibration probe** (tune_probe.ocn): 64 trim configurations $\times$ 5 corners, 320 transients; produces Fig. 7b and Table 6.
- **Calibrated campaign** (tdc_tuned_all.ocn): repeats the corner campaign with per-corner trim codes; produces Fig. 7a.
- **Metastability** (meta_sweep_{coarse,fine,ultra}.ocn): three nested sweeps, 49 transients, overdrive resolved to 0.5 fs; produces Fig. 9.
- **1-D comparison** (v1d_staircase_v2.ocn, v1d_energy2.ocn): 321-run staircase and energy measurement of the 1-D row; produces Fig. 10 and Table 5.